InlandEces Ltd.

Dublin, Ireland

System Administration Project

Linux | User & Group Management
Permissions, ACL & Security Hardening

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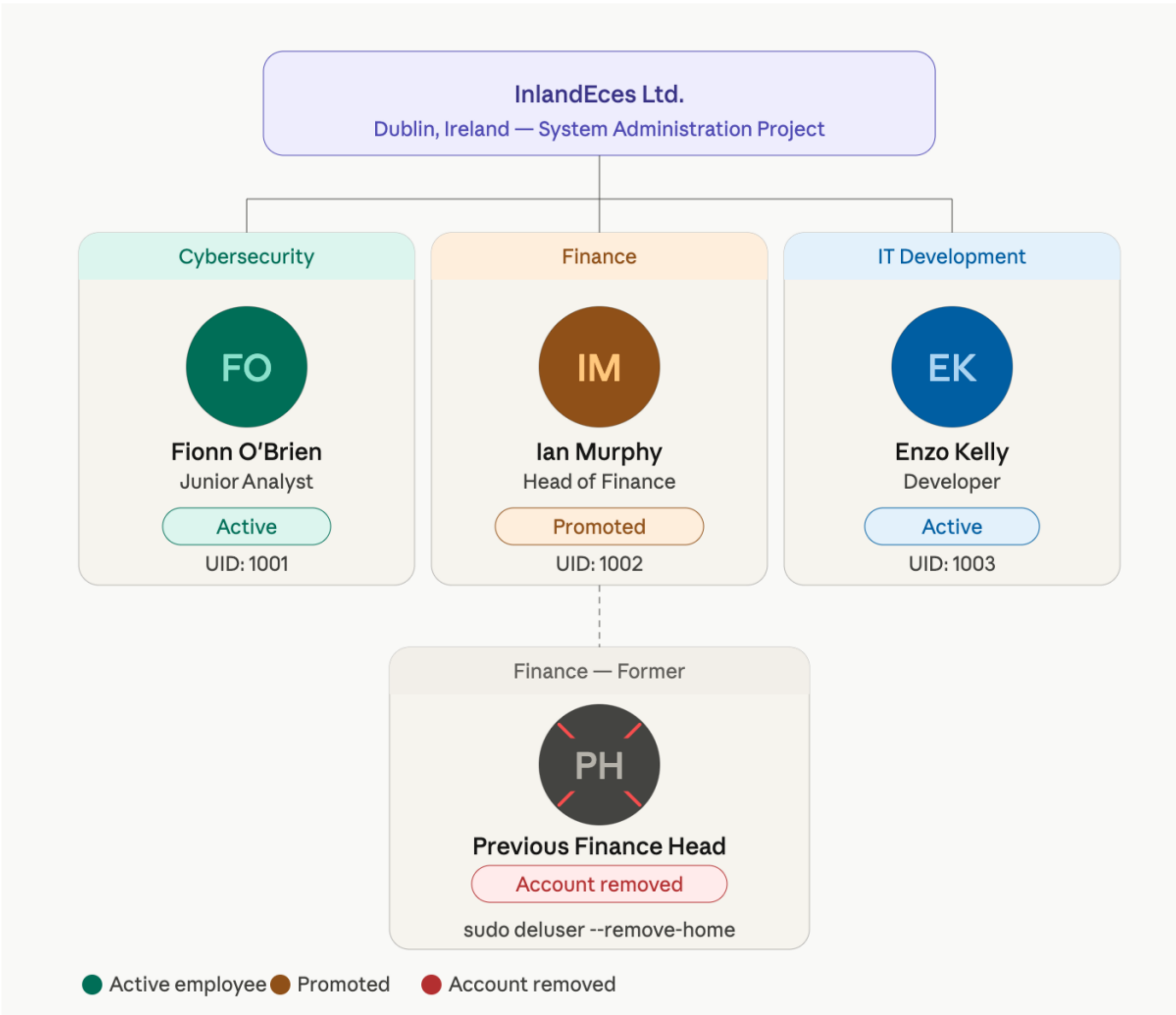
Environment	Kali Linux | Ubuntu | Windows Server

Purpose: This document demonstrates the manual, step-by-step process of setting up users, groups, directory structures, permissions, ACLs and security policies on a Linux system for a fictional Irish technology company. Every command shown was executed manually in the terminal to prove hands-on competence.

1. Company Scenario

InlandEces Ltd. is an Irish technology company based in Dublin. Following a period of team restructuring, three new employees need to be onboarded across different departments, and one department head position must be filled internally. The goal is to configure the Linux system manually — creating groups, users, directories and permissions from scratch — without relying on automation scripts.

Employee	Role / Action
Fionn O'Brien	Junior — Cybersecurity Team
Ian Murphy	Analyst → promoted to Head of Finance
Enzo Kelly	Developer — IT / Development Team
Previous Finance Head	Left the company — account to be removed



2. Linux Permissions — Theory

In Unix/Linux every file and directory has a set of permission bits that control who can read, write or execute it. Understanding this model is essential before touching any command.

2.1 Symbolic vs Octal Mode

Symbol	Octal	Meaning
r (read)	4	View file contents / list directory
w (write)	2	Modify file / create files in directory

x (execute)	1	Run file as program / enter directory

2.2 Permission String Anatomy

```
- rwx rwx rwx
│  │   │   └── Others (rest of the world)
│  │   └────── Group
│  └────────── Owner (user)
└──────────── File type  ( - file | d directory | l symlink )
```

2.3 Octal Quick Reference

Octal	Symbolic	Description
7	rwx	Full access — read, write, execute
6	rw-	Read and write — no execute
5	r-x	Read and execute — no write
4	r--	Read only
3	-wx	Write and execute — no read
0	---	No permissions at all

3. Step 1 — Inspect the Current System

Before creating anything, always audit what already exists on the machine.

Commands to run manually:

→[cat /etc/passwd]

View all system users

```
┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# ls
adduser.conf        doc-base      ipsec.secrets    motd      rc1.d     subversion
alsa                docker        issue            mtab      rc2.d     sudo.conf
alternatives        dpkg          issue.net        mysql     rc3.d     sudoers
```

```
┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# cat passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
```

→[awk -F: '$3 >= 1000 {print $1, $3}' /etc/passwd]
Show only real users (UID >= 1000)

```
┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# awk -F: '$3 ≥ 1000 {print $1, $3}' /etc/passwd
nobody 65534
red_team 1000
EnzoDiSotto 1001
```

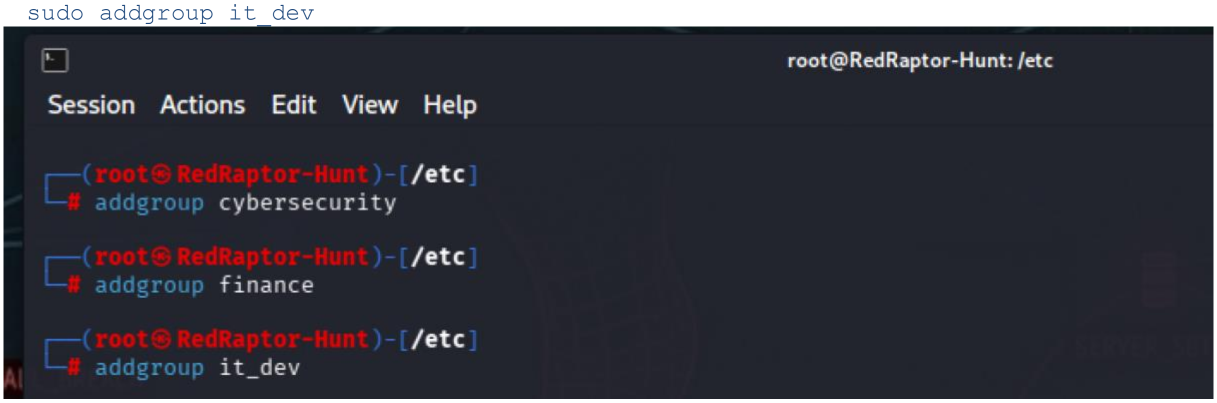
→[cat /etc/group]
View all groups and their members

```
┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# cat group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:red_team
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
```

4. Step 2 — Create the Department Groups

Groups are the backbone of permission management in Linux. Each department gets its own group before any users are created.

```
# Create the Cybersecurity group
sudo addgroup cybersecurity

# Create the Finance group
sudo addgroup finance

# Create the IT / Development group
sudo addgroup it_dev
```




```
# Verify all three groups were created
[grep -E 'cybersecurity|finance|it_dev' /etc/group]
```

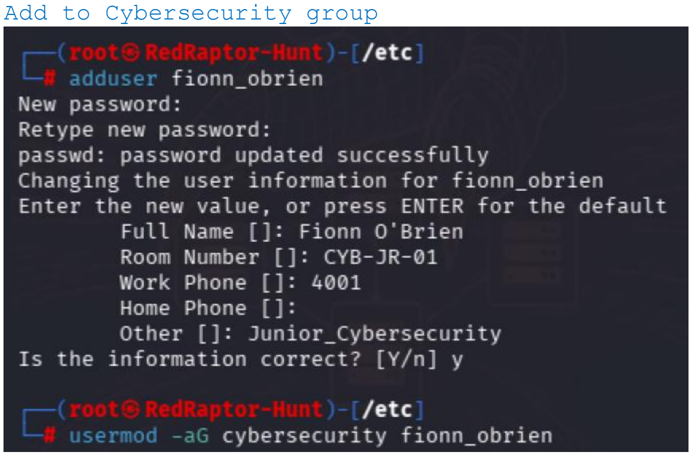
5. Step 3 — Create the Three Users Manually

Each user is created interactively with adduser. The system will prompt for a password and GECOS fields (full name, room, phone, etc.).

5.1 Fionn O'Brien — Junior Cybersecurity

```
→sudo adduser fionn_obrien
  Password:2768
  Full Name  : Fionn O'Brien
  Room       : CYB-JR-01
  Work Phone : 4001
  Other      : Junior_Cybersecurity
```

```
→[sudo usermod -aG cybersecurity fionn_obrien]
Add to Cybersecurity group
```

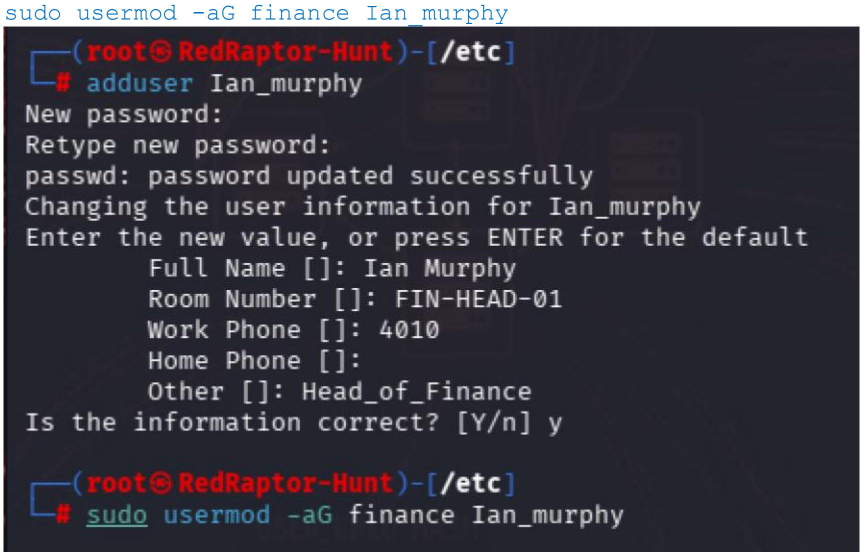


5.2 Ian Murphy — Head of Finance

```
→[sudo adduser Ian_murphy]
  Full Name  : Ian Murphy
  Password:2378
  Room       : FIN-HEAD-01
  Work Phone : 4010
  Other      : Head_of_Finance
```

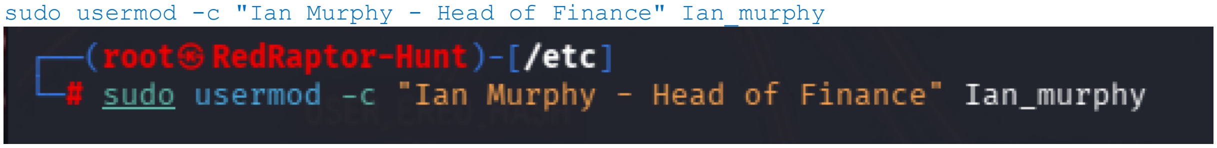
→**Add to Finance group**

```
sudo usermod -aG finance Ian_murphy
```



```
┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# adduser Ian_murphy
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for Ian_murphy
Enter the new value, or press ENTER for the default
        Full Name []: Ian Murphy
        Room Number []: FIN-HEAD-01
        Work Phone []: 4010
        Home Phone []:
        Other []: Head_of_Finance
Is the information correct? [Y/n] y

┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# sudo usermod -aG finance Ian_murphy
```

```
# Update GECOS to reflect her title as department head
sudo usermod -c "Ian Murphy - Head of Finance" Ian_murphy
```



```
┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# sudo usermod -c "Ian Murphy - Head of Finance" Ian_murphy
```

5.3 Enzo Kelly — IT Developer

→**sudo adduser Enzo_kelly**

```
  Full Name  : Enzo Kelly
  Password   : 8678
  Room       : IT-DEV-01
  Work Phone : 4002
  Other      : IT_Developer
```

```
# Add to IT Dev group
sudo usermod -aG it_dev Enzo_kelly
```

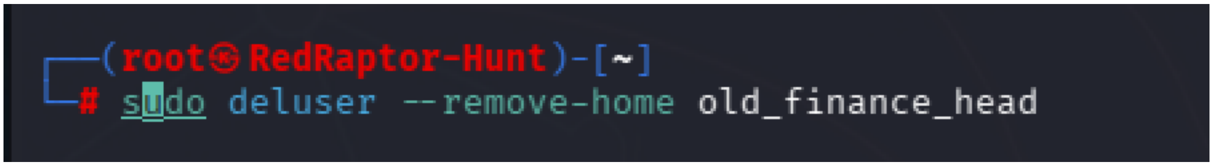
```
┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# sudo adduser Enzo_kelly
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
warn: `/bin/passwd Enzo_kelly' failed with status 10. Continuing.
warn: Wrong password given or retyped incorrectly
Changing the user information for Enzo_kelly
Enter the new value, or press ENTER for the default
        Full Name []: Enzo Kelly
        Room Number []: IT-DEV-01
        Work Phone []: 4002
        Home Phone []: IT_Developer
        Other []: IT_Developer
Is the information correct? [Y/n] n
Changing the user information for Enzo_kelly
Enter the new value, or press ENTER for the default
        Full Name [Enzo Kelly]: Enzo Kelly
        Room Number [IT-DEV-01]: IT-DEV-01
        Work Phone [4002]: 4002
        Home Phone [IT_Developer]:
        Other [IT_Developer]: IT_Developer
Is the information correct? [Y/n] y

┌──(root㉿RedRaptor-Hunt)-[/etc]
└─# sudo usermod -aG it_dev Enzo_kelly
```

> i **What is the UID?** Linux identifies users by number (UID), not by name. Real users start at UID 1000+. UID 0 is always root.

6. Step 4 — Remove the Previous Finance Head

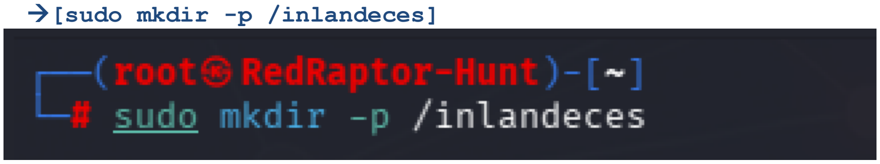
The previous Finance department head has left the company. The account must be properly removed from the system.

→ **`[sudo deluser --remove-home old_finance_head]`**
`Delete user AND their home folder (irreversible)`



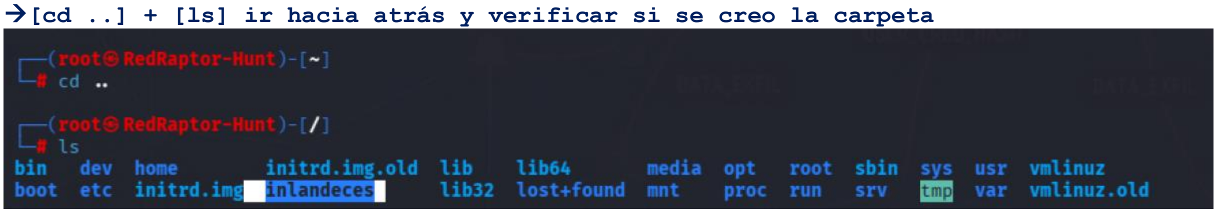
7. Step 5 — Create the Directory Structure

→[sudo mkdir -p /inlandeces]



Create root company directory

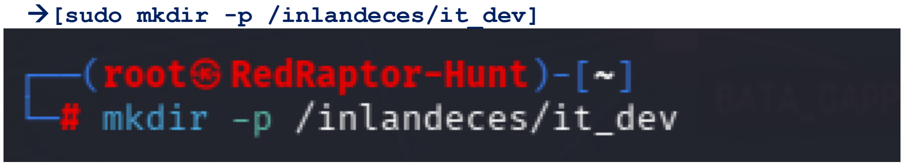
→[cd ..] + [ls] ir hacia atrás y verificar si se creo la carpeta



→[Create one subdirectory per department]
sudo mkdir -p /inlandeces/cybersecurity

→[sudo mkdir -p /inlandeces/finance]

→[sudo mkdir -p /inlandeces/it_dev]



→[cd ..] + [ls] ir hacia atrás y verificar si se creo la carpeta

```
┌──(root㉿RedRaptor-Hunt)-[/]
└─# cd inlandeces

┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# ls
cybersecurity  finance  it_dev
```

8. Step 6 — Assign Ownership and Permissions

Now we assign the correct owner and group to each directory, then set the permission bits with chmod.

8.1 chown — Transfer Ownership

→[sudo chown root:cybersecurity /inlandeces/cybersecurity]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# chown root:cybersecurity /inlandeces/cybersecurity
```

Cybersecurity dir → owned by root, group cybersecurity

→[sudo chown Ian_murphy:finance /inlandeces/finance]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# chown Ian_murphy:finance /inlandeces/finance
```

Finance dir → owned by Ian (dept head), group finance

→[sudo chown root:it_dev /inlandeces/it_dev]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# sudo chown root:it_dev /inlandeces/it_dev
```

8.2 chmod — Set Permission Bits

Directory	Mode	Explanation
/inlandeces/cybersecurity	750	Owner=rwx Group=r-x Others=---
/inlandeces/finanzas	750	Owner=rwx Group=r-x Others=---
/inlandeces/it_dev	770	Owner=rwx Group=rwx Others=---

→[ls -la] verificar los permisos que tiene.

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# ls -la
total 20
drwxr-xr-x  5 root root          4096 May 29 09:42 .
drwxr-xr-x 20 root root          4096 May 29 09:23 ..
drwxr-xr-x  2 root cybersecurity 4096 May 29 09:24 cybersecurity
drwxr-xr-x  2 root finance       4096 May 29 09:25 finance
drwxr-xr-x  2 root it_dev        4096 May 29 09:25 it_dev
```

→sudo chmod 750 /inlandeces/cybersecurity
→sudo chmod 750 /inlandeces/finance
→sudo chmod 770 /inlandeces/it_dev

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# chmod 750 /inlandeces/cybersecurity

┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# chmod 750 /inlandeces/finance

┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# chmod 770 /inlandeces/it_dev
```

→**[ls -la]** Verify all permissions at once

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# ls -la
total 20
drwxr-xr-x  5 root root          4096 May 29 09:42 .
drwxr-xr-x 20 root root          4096 May 29 09:23 ..
drwxr-x---  2 root cybersecurity 4096 May 29 09:24 cybersecurity
drwxr-x---  2 root finance       4096 May 29 09:25 finance
drwxrwx---  2 root it_dev        4096 May 29 09:25 it_dev
```

9. Step 7 — Fine-Grained Access with ACL

Standard Linux permissions apply the same rights to every member of a group. ACLs (Access Control Lists) allow per-user overrides within the same group — essential for distinguishing a Junior from a Senior in the same team.

9.1 Install ACL support

->[sudo apt update && sudo apt install acl -y]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# sudo apt install acl -y
acl is already the newest version (2.3.2-3).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 369
```

9.2 Apply ACL — Cybersecurity (Junior restriction)

→[sudo setfacl -m u:fionn_obrien:r-x /inlandeces/cybersecurity]

Fionn is a Junior — read and navigate only, cannot modify anything

→[getfacl /inlandeces/cybersecurity]

Inspect the result

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# setfacl -m u:fionn_obrien:r-x /inlandeces/cybersecurity

┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# getfacl /inlandeces/cybersecurity
getfacl: Removing leading '/' from absolute path names
# file: inlandeces/cybersecurity
# owner: root
# group: cybersecurity
user::rwx
user:fionn_obrien:r-x
group::r-x
mask::r-x
other::---
```

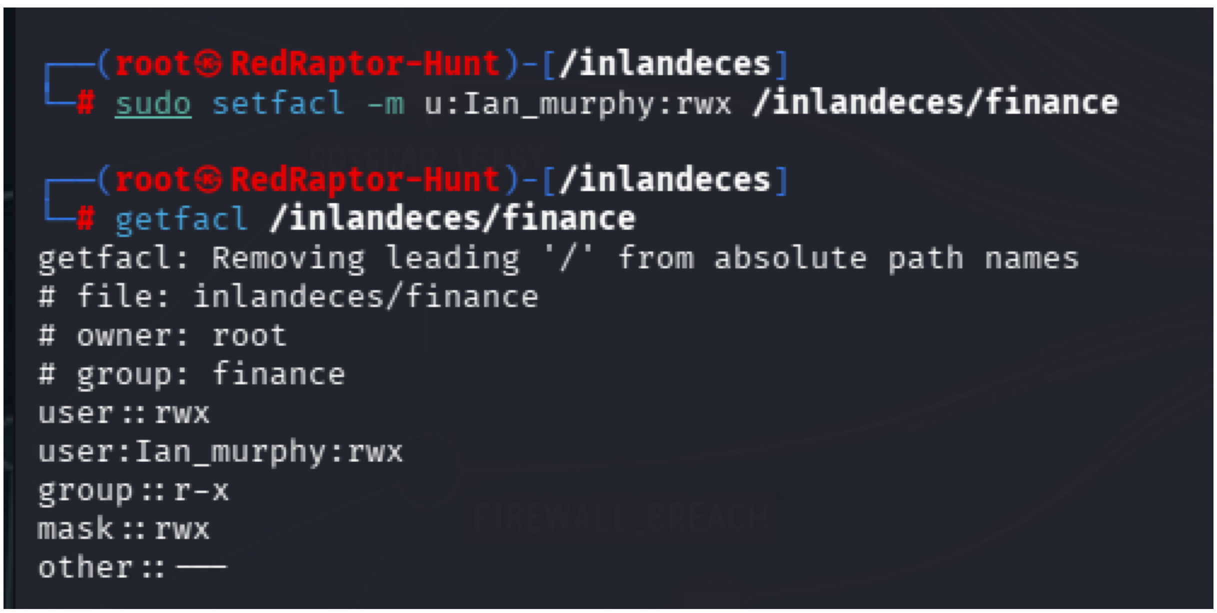
9.3 Apply ACL — Finance (Head full control)

```
#
```

→ **[sudo setfacl -m u:Ian_murphy:rwx /inlandeces/finance]**
Ian is Head of Finance — full control over her department folder

→ **[getfacl /inlandeces/finance]**

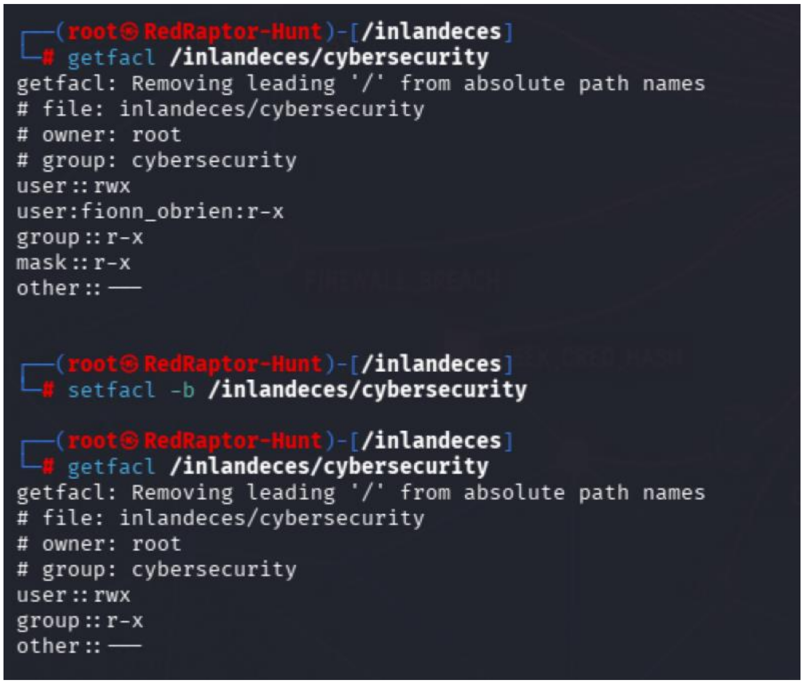
Inspect the result



9.4 Remove all ACL entries (reset)

→ **[getfacl /inlandeces/cybersecurity]**

Check who has access
→ **[sudo setfacl -b /inlandeces/cybersecurity]**



10. Step 8 — Security Hardening with umask

The umask defines which permission bits are blocked by default when a user creates a new file or directory. It does not add permissions — it subtracts them from the system base (777 for directories, 666 for files).

umask	New File	New Dir	Security Level
022	644 rw-r--r--	755 rwxr-xr-x	Standard — public read
027	640 rw-r-----	750 rwxr-x---	Private to group
077	600 rw-------	700 rwx------	Maximum security — owner only

10.1 Set umask permanently per user

```
→[sudo vim /home/fionn_obrien/.bashrc]
-Cybersecurity (Junior) — maximum privacy
-Add at the bottom:  umask 077
```

-->[cat /home/fionn_obrien/.bashrc]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# vim /home/fionn_obrien/.bashrc
```

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# cat /home/fionn_obrien/.bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

# If not running interactively, don't do anything
case $- in
```

```
# ═══════════════════════════════════════════════
# Cybersecurity (Junior) — maximum privacy
# ═══════════════════════════════════════════════
umask 077

┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─#
```

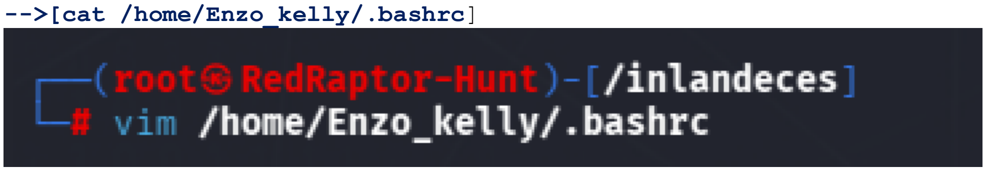
→**[sudo vim /home/Ian_murphy/.bashrc]** Finance Head — private but group-readable
→ Add at the bottom: umask 027
-->[cat /home/Ian_murphy/.bashrc]

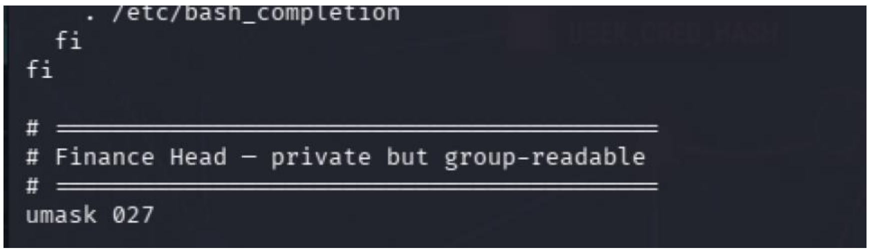
```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# vim /home/Ian_murphy/.bashrc
```

```
# ═══════════════════════════════════════════════
# Finance Head — private but group-readable
# ═══════════════════════════════════════════════
umask 027

┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─#
```

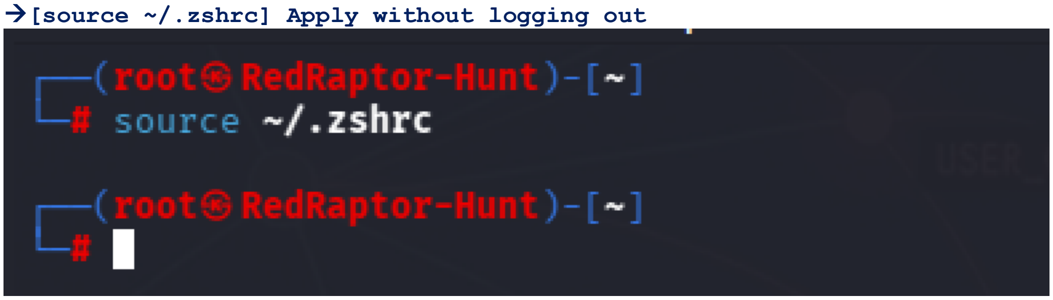
→ **[sudo vim /home/Enzo_kelly/.bashrc]** IT Developer — standard collaborative
-->Add at the bottom: umask 022

-->[cat /home/Enzo_kelly/.bashrc]





→[source ~/.zshrc] Apply without logging out



11. Step 9 — Hard Links and Symbolic Links

Linux supports two types of links that are useful for shared resources across departments.

11.1 Hard Link

→ [touch /inlandeces/finance/report.txt] Create the file -report.txt-
→[ls] Check if the file was created

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces]
└─# cd finance

┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# ls
report.txt
```

→[ln /inlandeces/finance/report.txt /inlandeces/finance/report_backup.txt]
Create a hard link

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# ln /inlandeces/finance/report.txt /inlandeces/finance/report_backup.txt
```

→[ls -li /inlandeces/finance/]
Both files point to the same inode (same data block)

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# ls -li /inlandeces/finance/
total 0
8519683 -rw-r--r-- 2 root root 0 May 29 11:50 report_backup.txt
8519683 -rw-r--r-- 2 root root 0 May 29 11:50 report.txt
```

→[rm /inlandeces/finance/report.txt]
Deleting one does NOT destroy the data — the other survives
→[ls -li]
Both files point to the same inode (same data block)

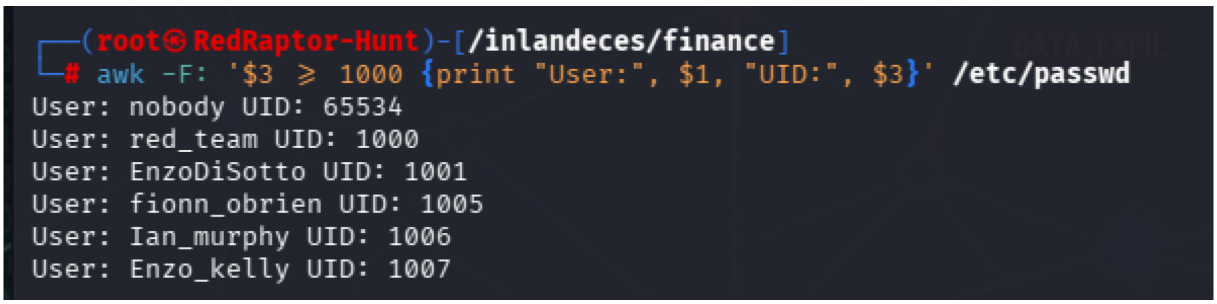
```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# rm /inlandeces/finance/report.txt

┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# ls -li /inlandeces/finance/
total 0
8519683 -rw-r--r-- 1 root root 0 May 29 11:50 report_backup.txt
```

12. Step 10 — Final System Verification

Run these commands after completing every step to confirm the full configuration is correct before sign-off.

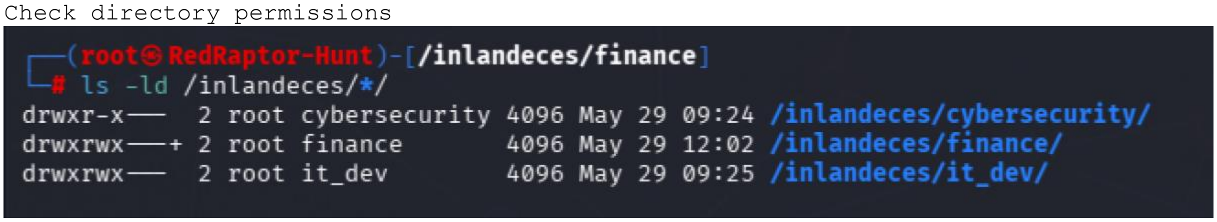
→ **[awk -F: '$3 >= 1000 {print "User:", $1, "UID:", $3}' /etc/passwd]**

List all real users



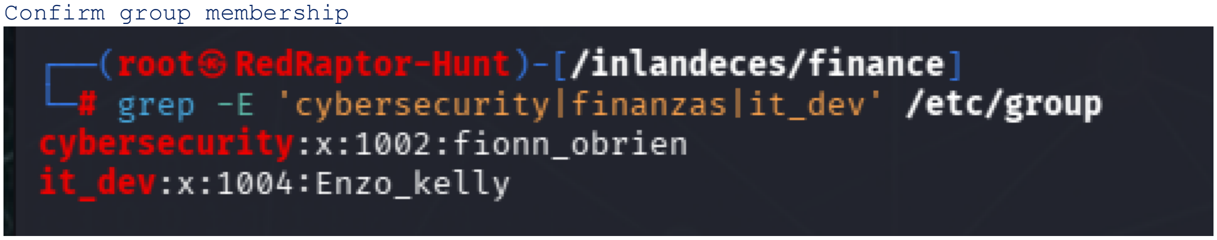
→ **[ls -ld /inlandeces/*/]**

Check directory permissions



→ **[grep -E 'cybersecurity|finanzas|it_dev' /etc/group]**

Confirm group membership



→ **[Review ACLs]**

-->[getfacl /inlandeces/cybersecurity]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# getfacl /inlandeces/cybersecurity
getfacl: Removing leading '/' from absolute path names
# file: inlandeces/cybersecurity
# owner: root
# group: cybersecurity
user::rwx
group::r-x
other::---
```

-->[getfacl /inlandeces/finance]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# getfacl /inlandeces/finance
getfacl: Removing leading '/' from absolute path names
# file: inlandeces/finance
# owner: root
# group: finance
user::rwx
user:Ian_murphy:rwx
group::r-x
mask::rwx
other::---
```

→[Confirm individual user groups]

-->[groups fionn_obrien]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# groups fionn_obrien
fionn_obrien : fionn_obrien users cybersecurity
```

-->[groups Ian_murphy]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# groups Ian_murphy
Ian_murphy : Ian_murphy users finance
```

-->[groups Enzo_kelly]

```
┌──(root㉿RedRaptor-Hunt)-[/inlandeces/finance]
└─# groups Enzo_kelly
Enzo_kelly : Enzo_kelly users it_dev
```

14. Appendix — Command Reference

Command	Description
ls -l file	View file permissions
ls -ld directory	View directory permissions
chmod 750 path	Set permissions in octal mode
chmod u+x path	Set permissions in symbolic mode
chown user path	Change file owner
chown user:group path	Change owner and group together
sudo addgroup name	Create a new group
sudo adduser name	Create a new user interactively
sudo usermod -aG grp usr	Add user to a group
sudo usermod -L user	Lock a user account
sudo usermod -U user	Unlock a user account
sudo deluser user	Delete user (keep files)
sudo deluser --remove-home user	Delete user and home folder
sudo groupdel group	Delete a group
getfacl path	View ACL permissions
sudo setfacl -m u:usr:rwx path	Apply ACL to a user
sudo setfacl -b path	Remove all ACLs from a path
umask	View current umask value
umask 077	Set umask (temporary — current session)